For first 5 questions
2.5 pts for correct answer
2.5 pts for explanation

CS556-C Fall 2023 Midterm

Name:

Closed book: no textbook, no electronic devices, one sheet of paper with handwritten notes.

Read carefully before answering! Write your answers on the white sheets given to you. Show all your work, not only the final answer.

Question 1 *(5 points)* If A and B are square and invertible, then AB is invertible, then $(AB)^{-1} = A^{-1}B^{-1}$.Explain.

False.
$(AB)^{-1} = B^{-1}A^{-1}$
And the latter is not equal to $A^{-1}B^{-1}$

Question 2 *(5 points)* If A is a 3-by-3 matrix and the equation $Ax = b$ has a unique solution, then A is invertible? Explain.

True. If the system has a unique solution, then there is no free variables, so A is invertible.

Question 3 *(5 points)* If a system of linear equations has two different solutions, it must have infinitely many solutions. True or False? Explain

True. Since a system can have 0, 1 or ∞ solutions
If it has 2, then it has an infinite numbers of solutions

Question 4 *(5 points)* If B, C, D are three matrices and $BC = BD$, then $C = D$. True or False?

False. $B(C-D)=0$ for C different from D.

Question 5 *(5 points)* The columns of an invertible m-by-m matrix form a basis for $\mathcal{R}^m$. True or False? Explain.

True. Columns are a set of m independent vectors which can form a basis for $\mathbb{R}^m$.

Question 6 *(25 points)* Let $x_1 = \begin{bmatrix}2\\3\\2\end{bmatrix}$, $x_2 = \begin{bmatrix}1\\2\\1\end{bmatrix}$, and $v = \begin{bmatrix}a\\b\\c\end{bmatrix}$.

a) Find a vector x_3 such that $\{x_1, x_2, x_3\}$ is a linearly independent set of vectors.

b) Find the determinant of the 3-by-3 matrix composed of columns x_1, x_2, x_3.

c) Find the rank of this matrix.

d) Find the inverse of this matrix.

e) Find a condition that a, b, c must satisfy for the set of the vectors $\{x_1, x_2, v\}$ to be linearly dependent.

6 pts
1 correct approach
2.5 for correct det
3 pts for rank, 0.5 for reasoning
4 for approach, 2 for correct answer
4 for approach, 2 for correct answer

a) In order for the set of vectors $\{x_1, x_2, x_3\}$ to be linearly independent the determinant of matrix $A = [x_1\, x_2\, x_3]$ should be different from 0.

Let $x_3 = \begin{bmatrix}1\\2\\0\end{bmatrix} \Rightarrow A = \begin{bmatrix}2&1&1\\3&2&2\\2&1&0\end{bmatrix}$

$\det(A) = 1\begin{vmatrix}3&2\\2&1\end{vmatrix} - 2\begin{vmatrix}2&1\\2&1\end{vmatrix} = -1 \neq 0$

Hence, the set of vectors is linearly independent.

b) $\det(A) = -1$, as found on part a).

c) $\text{rank}(A) = 3$ since the vectors are linearly independent.

d) Using Gauss-Jordan

$A^{-1} = \begin{bmatrix}2&-1&0\\-4&2&1\\1&0&-1\end{bmatrix}$

e) In order for the set to be linearly dependent $\Rightarrow \det(A) = 0$

$\begin{bmatrix}2&1&a\\3&2&b\\2&1&c\end{bmatrix} \Rightarrow \det(A) = a\begin{vmatrix}3&2\\2&1\end{vmatrix} - b\begin{vmatrix}2&1\\2&1\end{vmatrix} + c\begin{vmatrix}2&1\\3&2\end{vmatrix}$

$= -a + c$

$= c - a$

$\boxed{c - a = 0}$

Question 7 *(25 points)* Given matrix $B = \begin{bmatrix} 1 & 1 & 4 & 5 \\ 2 & 1 & 5 & 5 \\ 0 & 1 & 3 & 2 \\ -1 & 1 & 2 & 2 \end{bmatrix}$

a) Is there a non-trivial solution of the equation $Bx = 0$? 5 for row red, 5 for N(A)

b) Is there a vector b for which $Bx = b$ has no solutions? Write down such a vector if it exists. 5 for correct answer

c) Write down a vector c for which $Bx = c$ has at least one solution? Write down the complete solution of $Bx = c$. 5 for row red, 2 for x_p, 2 for x_n, 1 for X.

a) $\begin{bmatrix} 1 & 1 & 4 & 5 \\ 2 & 1 & 5 & 5 \\ 0 & 1 & 3 & 2 \\ -1 & 1 & 2 & 2 \end{bmatrix} \xrightarrow[R_4 = R_4 + R_1]{R_2 = R_2 - 2R_1} \begin{bmatrix} 1 & 1 & 4 & 5 \\ 0 & 1 & 3 & 5 \\ 0 & 1 & 3 & 2 \\ 0 & 2 & 6 & 7 \end{bmatrix} \xrightarrow[R_4 = R_4 - 2R_2]{R_3 = R_3 - R_2} \begin{bmatrix} \textcircled{1} & 1 & 4 & 5 \\ 0 & \textcircled{1} & 3 & 5 \\ 0 & 0 & 0 & -3 \\ 0 & 0 & 0 & \textcircled{-3} \\ x & y & z & w \end{bmatrix}$ $R_2 *= -1$

$w = 0$

$z = 1$, the free variable

$y = -3$

$x = -1$

$N(A) = \begin{bmatrix} -1 \\ -3 \\ 1 \\ 0 \end{bmatrix} s, \forall s \in \mathbb{R}.$

A non-trivial solution of the equation $Bx=0$ is $\begin{bmatrix} -1 \\ -3 \\ 1 \\ 0 \end{bmatrix}$

b) A vector for which the system has no solutions is $\begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$.

c) Let vector $c = \begin{bmatrix} 7 \\ 8 \\ 3 \\ 2 \end{bmatrix}$

$\left[\begin{array}{cccc|c} 1 & 1 & 4 & 5 & 7 \\ 2 & 1 & 5 & 5 & 8 \\ 0 & 1 & 3 & 2 & 3 \\ -1 & 1 & 2 & 2 & 2 \end{array}\right] \xrightarrow[R_4 = R_4 + R_1]{R2 = R2 - 2R1} \left[\begin{array}{cccc|c} 1 & 1 & 4 & 5 & 7 \\ 0 & 1 & 3 & 5 & 6 \\ 0 & 1 & 3 & 2 & 3 \\ 0 & 2 & 6 & 7 & 9 \end{array}\right] \xrightarrow[R_4 = R_4 - 2R_2]{R_3 = R_3 - R_2} \left[\begin{array}{cccc|c} 1 & 1 & 4 & 5 & 7 \\ 0 & 1 & 3 & 5 & 6 \\ 0 & 0 & 0 & -3 & -3 \\ 0 & 0 & 0 & 3 & 3 \end{array}\right]$ $R_2 *= -1$ $R_4 *= -1$

$\xrightarrow[R_3 /= -3]{R_4 = R_4 + R_3} \left[\begin{array}{cccc|c} \textcircled{1} & 1 & 4 & 5 & 7 \\ 0 & \textcircled{1} & 3 & 5 & 6 \\ 0 & 0 & 0 & \textcircled{1} & 1 \\ 0 & 0 & 0 & 0 & 0 \\ x & y & z & w & \end{array}\right]$

$x_{particular} = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$

$X = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ -3 \\ 1 \\ 0 \end{bmatrix} s, \forall s \in \mathbb{R}$

Question 8 *(25 points)* Consider the following system of linear equation for some constant k:

$$x + 2y - z = -2$$
$$5x - 2y + z = 6$$
$$-3x + 6y - 3z = k$$

a) For which value(s) of k does this system have one or infinitely many solutions?

b) Find all solutions for each value of k.

a) 7.5 for row red
5 for correct k
b) 5 for x_p
5 for x_n
2.5 for X

a) $\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 5 & -2 & 1 & 6 \\ -3 & 6 & -3 & k \end{array}\right] \xrightarrow[R3=R3+3R1]{R2=R2-5R1} \left[\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & -12 & 6 & 16 \\ 0 & 12 & -6 & k-6 \end{array}\right] \xrightarrow{R_3=R_3+R_2}$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & -2 \\ 0 & -12 & 6 & 16 \\ 0 & 0 & 0 & k+10 \\ x & y & z & \end{array}\right]$

System has infinitely many solution when $k = -10$

b) $x_{particular}$

$z = 0$

$y = -4/3$

$x = 2/3$

$x_{particular} = \begin{bmatrix} 2/3 \\ -4/3 \\ 0 \end{bmatrix}$

$x_{nullspace}$

$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 0 & -12 & 6 & 0 \\ 0 & 0 & 0 & 0 \\ x & y & z & \end{array}\right]$

$z = 1$

$y = 1/2$

$x = 0$

$x_{complete} = \begin{bmatrix} 2/3 \\ -4/3 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1/2 \\ 1 \end{bmatrix} s, \forall s \in \mathbb{R}.$

Question 9 *(5 points)* [Bonus] Please take a few minutes to share your thoughts on the course so far. Please be thoughtful and frank. Thank you.

- What do you like about the course so far?
- What suggestions do you have to improve this course?
- Do you have any other comments?